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|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------|
| 6(a)     | <b>response</b>                                                                                                                                                                                                                                                                                                                                          | <b>explanation</b>                                                  | <b>4</b> |
|          | blood vessels near the surface of the skin constrict (narrow)                                                                                                                                                                                                                                                                                            | less heat lost as, less / slower, blood (flow) at, skin / surface ; |          |
|          | shivering occurs                                                                                                                                                                                                                                                                                                                                         | <u>muscle contraction</u> , creates / generates / produces, heat ;  |          |
|          | hairs rise                                                                                                                                                                                                                                                                                                                                               | trapped air, insulates / reduces thermal gradient ;                 |          |
|          | reduction in the production of sweat                                                                                                                                                                                                                                                                                                                     | to stop / decrease, heat loss by <u>evaporation</u> ;               |          |
| 6(b)     | decrease / less, surface area to, conserve / retain heat<br><b>or</b><br>decrease / less, surface area so less heat lost ;                                                                                                                                                                                                                               |                                                                     | <b>1</b> |
| 6(c)(i)  | any <b>three</b> from:<br><b>1</b> Sitatunga's (core / body) temperature (fairly) constant ;<br><b>2</b> Nile monitor's / lizard's, (core / body) temperature rises as environmental temperature rises ;<br><b>3</b> figures for two animals at one temperature with units<br><b>or</b><br>one animal at two temperatures with units ;<br><b>4</b> AVP ; |                                                                     | <b>3</b> |
| 6(c)(ii) | problem with, catching prey / feeding / avoiding predation ;                                                                                                                                                                                                                                                                                             |                                                                     | <b>1</b> |
| 6(d)     | <b>1</b> <i>ref. to</i> set point / norm(al) / ideal value ;<br><b>2</b> rise / fall / change / fluctuation ;<br><b>3</b> reversed / corrected / counter-acted ;                                                                                                                                                                                         |                                                                     | <b>3</b> |